Direct Proportion

Exam code: 0607

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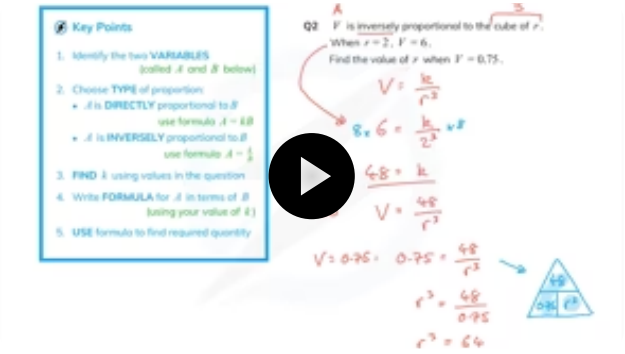

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Direct proportion

What is direct proportion?

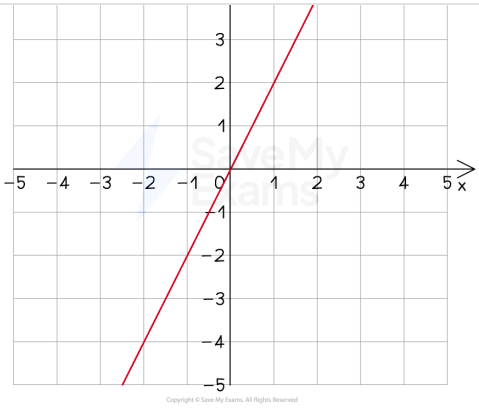
- **Proportion** is a way of talking about how two **variables** are related to each other
- **Direct** proportion means that as one variable goes **up** the other goes up by the same **factor**
  - The **ratio** between the two amounts will always stay the **same**
- The symbol $\propto$ means "proportional to"
  - E.g. y is directly proportional to x, $y \propto x$
- If x and y are directly proportional, then
  - $x : y$ will always be the same
  - there will be some value, k, such that $\boldsymbol{y = kx}$
  - the graph relating x and y is a linear graph, with **gradient** $\boldsymbol{k}$
- k is called the **constant of proportionality**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Example of a graph showing direct proportion

How do I use direct proportion with powers and roots?

- Problems may involve a variable being **directly proportional** to a **power or root** of another variable
- For example
  - *y* is directly proportional to the **square of *x***
    - $y \propto x^2$
    - means that $y = kx^2$
  - *y* is directly proportional to the **square root of *x***
    - $y \propto \sqrt{x}$
    - means that $y = k\sqrt{x}$
  - *y* is directly proportional to the **cube of *x***
    - $y \propto x^3$
    - means that $y = kx^3$
  - *y* is directly proportional to the **cube root of *x***
    - $y \propto \sqrt[3]{x}$
    - means that $y = k\sqrt[3]{x}$
  - Each of these would have a **different type of graph**, depending on the power or root

How do I find the equation between two directly proportional variables?

- Direct proportion questions always have the same process:
  - **STEP 1**
    Identify the two variables and write down the **formula in terms of *k***
    - E.g. ***y*** is **directly** proportional to ***x***
    - write down the formula $y = kx$
  - **STEP 2**
    Find *k* by substituting any given values **from the question** into your formula, then solving to get *k*
    - E.g. if you are told *y* = 6 when *x* = 2
    - then $6 = k \times 2$ giving $k = 3$
  - **STEP 3**
    Rewrite the formula with the **known value of *k*** from above (substitute it in)

Use the equation to answer other parts of the question

- E.g. find y when $x = 10$
- $y = 3x$ gives $y = 3 \times 10 = 30$

Examiner Tips and Tricks

Some harder exam questions do not tell you to work out the equation. You are expected to do it on your own.

Worked Example

It is known that y is directly proportional to the square of x.

When $x = 3$, $y = 18$.

Find the value of y when $x = 4$.

Answer:

Identify the two variables

$$y, x^2$$

We are told this is **direct** proportion
Write down the formula involving k

$$y = kx^2$$

Find k by substituting in $y = 18$ when $x = 3$
Then solve the equation for k

$$18 = k(3)^2$$
$$18 = 9k$$
$$\frac{18}{9} = k$$
$$2 = k$$

Substitute this value of k back into the formula to get the full equation

$$y = 2x^2$$

Use this formula to find y when $x = 4$

$$y = 2 \times 4^2$$
$$y = 2 \times 16$$
$$y = 32$$

$$y = 32$$

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